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A Forest Fire Prediction Framework Based on Multiple Machine Learning Models

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Abstract: Fire risk prediction is of great importance for fire prevention. Fire risk maps are an effective tool to quantify regional fire risk. Most existing studies on forest fire risk maps mainly use a single machine learning model, but different models have varying degrees of feature extraction in the same spatial environment, leading to inconsistencies in prediction accuracy. To address this issue, this study proposes a novel integrated machine learning framework that systematically evaluates multiple models and combines their outputs through a weighted ensemble approach, thereby enhancing prediction robustness. During the feature selection stage, factors including socio-economic, climate, terrain, remote sensing data, and human factors were considered. Unlike previous studies that mainly use a single model, eight models were evaluated and compared using performance metrics. Three models were weighted based on Mean Squared Error (MSE) values, and cross-validation results showed an improvement in model performance. The integrated model achieved an accuracy of 0.8602, an area under the curve (AUC) of 0.772, and superior sensitivity (0.9234), outperforming individual models. Finally, the weighted framework was applied to generate a fire risk map. Compared with prior studies, this multi-model ensemble approach not only improves predictive accuracy but also provides a scalable and adaptable framework for fire risk mapping, and provides valuable insights to address future fire sustainability issues.

Keywords: fire risk zones; forest fire prediction; forest fire influencing factors; machine learning

Academic Editor: Giorgio Vacchiano

Received: 20 January 2025
Revised: 10 February 2025
Accepted: 11 February 2025
Published: 13 February 2025

Citation: Wang, C.; Liu, H.; Xu, Y.; Zhang, F. A Forest Fire Prediction Framework Based on Multiple Machine Learning Models. *Forests* **2025**, *16*, 329. https://doi.org/10.3390/f16020329

1. Introduction

Forest ecosystems provide both direct and indirect economic values to humans, while playing a fundamental and strategic role in maintaining the Earth's ecological balance and promoting sustainable development. However, forest fires represent one of the most destructive natural disasters, causing severe damage to both the ecosystem and human property [1]. These fires can lead to the loss of biodiversity, disrupt carbon storage, and contribute significantly to air pollution, including the release of greenhouse gases [2]. The destruction of forests not only harms wildlife but also compromises the essential services they provide, such as regulating water cycles and preventing soil erosion [3].

The causes of forest fires are very complex [4], involving various topographic factors, climate factors, socio-economic factors, and human factors [5]. Terrain is the main factor affecting fire ignition, as it can affect the distribution of a local climate (including sunlight, temperature, etc.). Affected by convective and radiative heat, the fire may spread rapidly along steep and upward slopes, but slowly in areas sloping downwards. Wind speed can affect the dryness of combustibles, which in turn affects the occurrence of fires. Due to differences in temperature and vegetation, the probability of fires occurring in different altitude areas may vary. The TWI (Terrain Moisture Index), to some extent, indicates the influence of terrain and soil characteristics on soil moisture distribution [6]. Forest fires are influenced by hydrological conditions. Recently, TWI and NDVI (Normalized Difference Vegetation Index) have been found to be effective factors in predicting the occurrence of fires [7]. Slope orientation, altitude, gradient, and TWI can be extracted from digital elevation model (DEM) data [8].

Climate factors including temperature, wind speed, air pressure, and rainfall are important triggers for fire ignition [9]. High temperatures can lead to low soil moisture and even drought, making forests prone to fires [10]. NDVI is widely chosen to represent vegetation distribution, which is often considered a representative for analyzing fuel used in fire incidents [11].

Accurate fire risk maps can help fire agencies allocate fire prevention resources throughout the forest in advance and take emergency measures to control the fire in a small area [12]. In this way, more forest resources and human property can be better protected. The fire risk map is based on the geographic information of the study area, combined with advanced tools and methods, which has a great impact on fire prevention [13].

In recent years, forest fires in China have caused various types of damage, including economic losses, threats to human safety, environmental degradation, and social impacts. The burning area of global forest fires in 2024 was about 3.9 million hectares [14]. China's total forest area is 231 million hectares [15]. The impacts caused by forest fires in recent years primarily include the following aspects: In terms of human casualties, forest fires have occurred frequently across China, leading to severe loss of life. For example, since the spring of 2019, major forest fires in provinces including Shanxi, Sichuan, Inner Mongolia, and Hebei have resulted in significant casualties [16].

In terms of economic losses, forest fires cause not only direct economic losses, including the destruction of timber resources and firefighting costs, but also indirect economic losses, including production suspension, reduced output, and losses in tourism and tax revenues. For example, between 2000 and 2014, the cumulative economic loss from forest fires in China amounted to CNY 80.93 million [17]. In terms of ecological impact, forest fires have severely affected the environment, causing the destruction of forest soils, air quality, and natural landscapes, as well as impacting plant and animal resources. Additionally, forest fires lead to the loss of carbon sinks. For instance, between 1996 and 2013, the direct and indirect carbon sink losses caused by forest fires in China totaled 67.52 million tons [18]. In terms of social harm, forest fires also have a wide range of social impacts, including threats to social stability and public safety, as well as adverse effects on the lives of residents [19].

Moreover, between 2001 and 2016, Anhui Province experienced 2336 forest fires, with an average of 146 fires per year [20]. The average area burned annually was 785.9 hectares, with 331.8 hectares of forest affected, resulting in an average loss of 7239.1 cubic meters of forest timber each year. Although the frequency of fires has declined year by year, the risk of forest fires remains significant. Notably, approximately 90% of forest fires in Anhui are caused by human activities. Among the identified fire sources, 569 fires (24.4%) were caused by productive activities, while 1,476 fires (63.2%) were due to non-productive activities. Among the productive fire sources, the top 3 causes were slash-and-burn farming (17.1%), forest regeneration through burning (4.2%), and kiln burning (0.5%). Among non-productive fire sources, the most common causes were burning offerings at graves (44.7%), outdoor smoking (5.3%), and children playing with fire (4.2%). Therefore, considering the influence of residential and industrial areas is crucial for forest fire risk studies. Special attention should be given to fire source management and prevention measures in these areas to effectively reduce the likelihood of fire occurrences and mitigate the damage caused by fires.

In recent years, embedded machine learning models and supervised machine learning models have been commonly used in the study of forest fires [7]. Embedded machine learning models can process audio and image inputs on low-power IoT devices, enabling real-time detection of forest fires. The advantage of this approach is its ability to achieve efficient fire detection in resource-constrained environments while using long-range wide area network (LoRaWAN) protocols to remotely signal events, ensuring timely responses [21]. Support Vector Machine (SVM) is a supervised learning method with good stability, which is widely used for the accurate recognition and classification of fires and smoke [22].

Recently, deep learning techniques have made significant strides in fire detection and prediction, with numerous studies exploring how to leverage advanced neural network models to enhance forest fire monitoring and early warning capabilities. A deep learning approach was proposed by [23], which integrates visible light imagery, infrared imagery, and the Forest Fire Weather Index (FWI), utilizing the YOLOv5 model and incorporating the EMA attention mechanism to optimize feature extraction, thereby improving the accuracy of smoke detection. This method effectively reduced false positives caused by non-fire factors such as clouds and dust, enhancing the reliability of early forest fire detection. However, this approach relies on pre-trained models, which may have limited adaptability to novel fire patterns, and its ability to control false positives in the face of extreme weather conditions or complex terrain environments remains to be further optimized. A deep learning-based time-series analysis method was proposed by [24], which is based on Long Short-Term Memory (LSTM) networks, combined with an improved Forest Fire Danger Index (FFDI) and covariate-assisted spatial interpolation techniques, successfully constructing a fire risk assessment framework with high spatiotemporal resolution. This method not only helped reconstruct historical forest fire risks from the pre-satellite era but also provided an important basis for future fire prediction. However, since this method primarily relies on historical data for training, the model's adaptability may be limited when fire patterns change significantly or are heavily influenced by climate change. Furthermore, the accuracy of this framework largely depends on the quality of the input data; insufficient data coverage or biases may affect the reliability of the prediction results. A CNN-BiLSTM model, combining Convolutional Neural Networks (CNNs) with Bidirectional Long Short-Term Memory (BiLSTM), has also been proposed to achieve near-real-time wildfire spread prediction [25]. This method utilizes VIIRS (Visible Infrared Imaging Radiometer Suite) active fire data, along with various environmental variables such as terrain, Normalized Difference Vegetation Index (NDVI), temperature, wind speed, and precipitation, effectively improving the timeliness of fire spread prediction. However, the performance of this model is still affected by the burn map generation method and may be reduced by the presence of false-positive (FP) and false-negative (FN) pixels. In practical applications, accurately obtaining burn rate information is crucial for optimizing spread prediction; if such information is missing, it may be necessary to rely on historical wildfire data for estimation, which could affect the model's generalization ability.

A supervised machine learning model is used to detect fire situations and burned areas. This method achieves high-accuracy fire detection by using multispectral Sentinel-2 images as input and combining various supervised classification techniques [26]. The RF model is also widely used to analyze the probability and driving factors of grassland fires and calculate the risk level of grassland fires [27]. A BP neural network can automatically extract patterns between output data through learning and adaptively memorize the learning content in the network's weights; with high self-learning and adaptive capabilities, it is applied to forest fire monitoring, combined with digital image processing technology [28]. XGBoost is used to predict the area of forest fire damage, which can accurately predict the damaged area of a fire [29]. CatBoost is an efficient gradient boosting algorithm primarily used for processing classification features and large-scale datasets, which is used to construct a forest fire prediction model for Guangdong Province [30]. KELM is an improved extreme learning machine algorithm based on kernel functions. In practical applications, KELM has been widely used for various prediction and classification tasks, which is used for forest fire image recognition [31].

Although a single machine learning model has shown good performance in forest fire prediction in specific areas, different models may have performance differences in predicting in the same area, and different models have varying degrees of feature extraction in the same spatial environment. Therefore, it is crucial to evaluate the differences in model performance and integrate the results of different models. This paper proposes a new ML integrated framework for drawing, evaluating, and predicting fire risk areas. In the feature selection stage, factors including socio-economic, climate, terrain, remote sensing data, and human factors were considered. After selecting potential features, Pearson correlation analysis and multicollinearity analysis were conducted to finalize 10 key features. This paper analyzes the factors that affect the occurrence of forest fires in the study area; evaluates the results of eight different models predicting forest fires, including including an SVM [32] model optimized using PSO [33], RF [34], CPO-BP [35], GOOSE-XGBoost [36], HLOA-CatBoost [37], HO-BP [38], BFO-KELM [39], and PO-BP [40]; and integrates the results of different models using weighting. The AUC and accuracy values obtained through cross-validation indicate that the weighted performance is better than the originals. The obtained fire risk framework can be used to draw a fire risk map.

At present, there is no machine learning-based fire risk area mapping research in the Yongqiao District in Anhui Province. Therefore, it is crucial to comprehensively examine the ability of mixed/single ML models to elucidate current and future forest fire patterns in the study area.

This paper thoroughly analyzed these key indicators, providing data-driven insights for local forest fire management, promoting data-based decision-making processes, and offering a scientific basis for fire prevention and management. Therefore, the main contributions of this paper are as follows:

(1) This paper analyzes the factors causing forest fires in the study area and evaluates the results of different models predicting forest fires.
(2) This paper uses comprehensive weighting to integrate the results of different models into a framework and verifies it using cross-validation.
(3) A new fire risk map was generated for the study area, which is beneficial for the allocation of fire prevention resources.

2. Materials and Methods

2.1. Study Area

The vegetation zoning of the Yongqiao District belongs to the warm temperate deciduous broad-leaf forest belt, and it belongs to the warm temperate semi-humid monsoon agricultural climate zone, with the characteristics of a transitional climate type from north to south. It is located in the northern part of the study area, under the jurisdiction of Suzhou City. It is situated between 33.33 N to 34.03 N latitude and 169.98 E to 170.33 E, with distinct four seasons, large temperature differences between day and night, sufficient sunshine, and moderate rainfall, and the dominant wind direction throughout the year is easterly, with an average annual wind speed of 2.6 m/s. Due to its location in the transitional zone between the north and south climates, winter is dry and summer is hot and rainy, with distinct four seasons, large temperature difference between day and night, sufficient sunshine, and moderate rainfall.

The vegetation of the area is primarily composed of warm temperate deciduous broadleaf forests, including fruit forests and mixed forests of trees and shrubs. Tree species in the region include poplars, cypress, locusts, peach trees, and apricot trees. Figure 1 shows the geographic location of the study area.

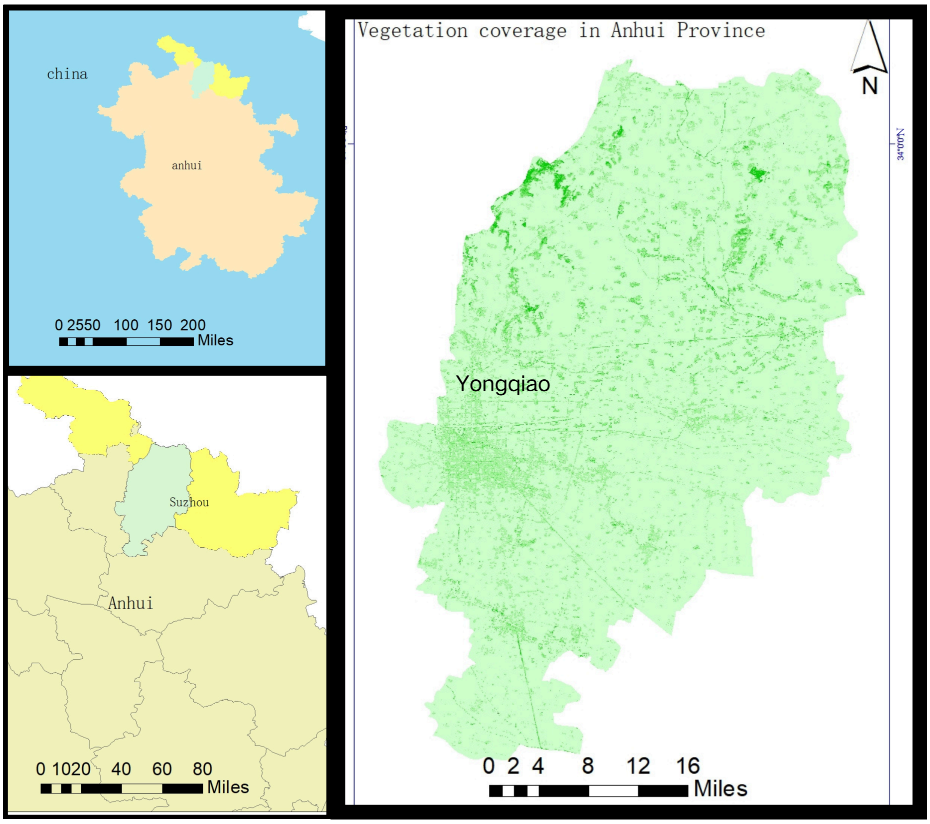


Figure 1. The geographic location of the study area.

2.2. Data Used

The source of the data used in this paper is shown in Table 1. We obtain the different data in different sources. Climate factors include TD (daily temperature), HD (dew point temperature), WSD (average wind speed), ARP (average air pressure), and rainfall. Topographic factors include altitude, TWI, aspect, slope, and NDVI (vegetation cover is used to display the geographical location of the study area, not as training data). Socioeconomic factors include distance to industrial areas, distance to road, and distance to river. Humanistic factors include distance to residential areas and population density.

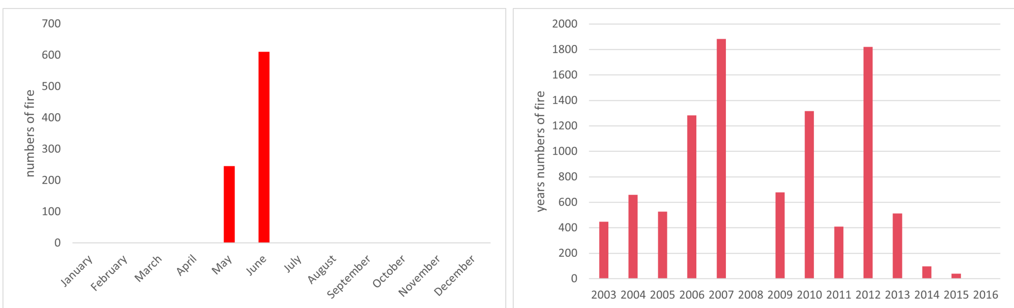
The overall altitude difference in the study area is not significant, with only slightly higher elevations in the northeast and northwest regions. The TWI values in the study area show an overall trend of low values in the north and high values in the south. Due to the high terrain in the southwest and low terrain in the northeast of the study area, the difference in aspect and slope values is not significant. Due to the significant difference in NDVI values in the study area, the difference between the maximum and minimum values is 0.871, showing a situation of low values in the south and high values in the north. The industrial and residential areas are located very close to each other, and the population density distribution and the distribution of residential and industrial areas show a similar trend but not completely the same. Roads are densely distributed in areas with high population density and sparsely distributed in areas with low population density, and the distribution of rivers within the study area is relatively uniform (Supplementary Figure S2).

Table 1. Data source.

Data Type	Name	Source
Climate Factors	TD HD ARP WSD Rainfall	(NOAA) https://www.ncei.noaa.gov/data/global-summary-of-the-day (accessed on 1 October 2024)
Topographic Factors	Altitude TWI Slope Aspect	https://www.gscloud.cn/sources/ (accessed on 1 October 2024)
	NDVI Vegetation cover	Google Earth Engine (GEE) platform
Socio-economic Factor	Distance to industrial areas (Dis 2) Distance to road Distance to river	EULUC-China https://www.openstreetmap.org/ (accessed on 2 October 2024)
Humanistic Factors	Distance to residential areas (Dis 1) Population density	EULUC-China ORNL LandScan Viewer - Oak Ridge National Laboratory
Forest fire data		https://daac.ornl.gov/ (accessed on 2 October 2024)

June is the season with the highest number of fires, so June 2015 is used as an example of climate factors (Figure 2). Although it can be seen from Supplementary Figure S2 that some factors have a small range within the time period shown, in fact, they have a large range throughout the year, as shown in Table 2.

The rainfall in the study area gradually increases from south to north. The difference between ARP and HD values in the study area is not significant; in June, the HD value was around 17 degrees Celsius and the ARP value was around 1000 hpa. The TD value of the study area is relatively low in areas with high vegetation coverage and relatively high in areas with low vegetation coverage. The difference in WSD values within the study area is minimal (Supplementary Figure S1).



(a) **(b)**

Figure 2. The monthly (**a**) and annual (**b**) distribution of forest fires.

Table 2. Climate factors.

Data Type	Name	Range
Climate Factors	TD	−6.2222 to 32.1667 (Celsius)
	HD	−16.44 to 25.6777 (Celsius)
	ARP	993.9 to 1033 (hpa)
	WSD	0.97444 to 5.24733 (m/s)
	Rainfall	6.442017 to 66.87158 (mm)

This paper collects forest fire data from 2003 to 2016. During this period, there were a total of 857 forest fire records in the study area from the Global Fire Atlas. It is noted that the majority of forest fires occur in May and June, suggesting that climatic and dry conditions during this period are key factors contributing to the occurrence of fires.

2.3. Method

The framework follows a five-stage workflow (Figure 3). (a) This paper grids the study area by 15 m × 15 m cells and uses the ArcGIS platform to generate layers for drawing fire ignition points and non-fire locations, and then selects various possible influencing factors. (b) Collect data and divide them into a test set and a training set. (c) Perform correlation analysis and multicollinearity testing on factors; select appropriate factors to proceed to the next stage of use. (d) Several ML algorithms, namely, PSO-SVM, RF, HO-BP, PO-BP, CPO-BP, GOOSE-XGBoost, HLOA-CatBoost, and BFO-KELM, were applied to predict the final fire risk warning map. After training the machine learning models, evaluate the performance using evaluation metrics including accuracy and AUC. (e) Weighted models with good performance are used to generate a fire risk map.

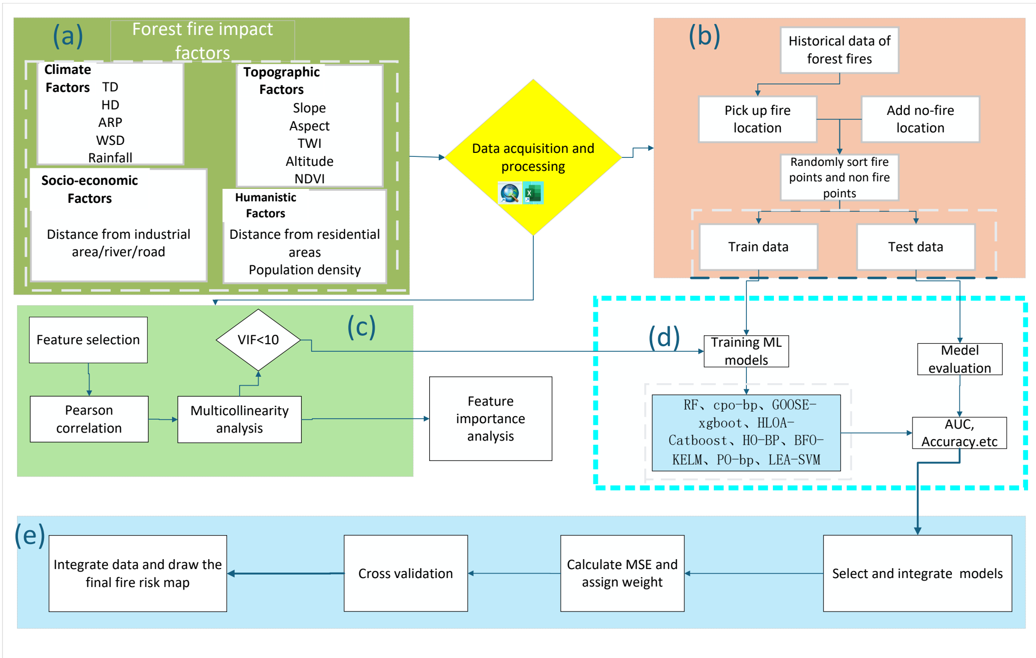


Figure 3. Detailed workflow of the method. (**a**) Factors selection. (**b**) Data processing. (**c**) Data check. (**d**) Models selection. (**e**) Generate map.

2.3.1. Correlation Analysis and Multicollinearity Detection Results

To determine the optimal features for the model, the importance of the independent variables was analyzed using Pearson correlation analysis and multicollinearity. The Pearson correlation coefficient is a ubiquitous statistical tool, extensively employed across diverse fields requiring the assessment of linear relationships between variables, such as the social sciences, natural sciences, medicine, and economics [41,42]. It is utilized for detecting and addressing multicollinearity issues, ensuring that redundant variables do not negatively impact model performance. Generally speaking, a Pearson correlation coefficient absolute value between 0.8 and 1 indicates a high correlation, that between 0.5 and 0.8 suggests a moderate correlation, that between 0.3 and 0.5 indicates a weak correlation, and that below 0.3 reflects a very weak or negligible correlation, which can be considered not correlated. After performing the correlation and multicollinearity analysis, unsuitable independent variables were removed. During this process, rainfall was highly correlated with TD, distance to road and distance to river were highly correlated with population density, and slope and aspect were highly correlated with TWI. Their Pearson correlation coefficients all exceeded 0.8; therefore, rainfall, slope, aspect, distance to road, and distance to river were removed. The final selection consisted of the following 10 features: altitude, TWI, NDVI, TD, HD, ARP, WSD, population density, distance to residential areas, and distance to industrial areas. Supplementary Figure S3 shows the Pearson correlation analysis of the 10 selected features.

Variance Inflation Factor (VIF) is used to evaluate the degree of multicollinearity between related variables (Supplementary Table S1). A VIF close to 1 indicates that there is almost no multicollinearity, a VIF greater than 10 is generally considered to be the threshold for multicollinearity, and a VIF greater than 20 indicates a serious multicollinearity problem. In this study, variables with VIFs below 10 were considered acceptable for inclusion. For example, despite potential interactions, the variables TWI (VIF = 1.0013) and NDVI (VIF = 1.045) were retained due to their distinct roles in capturing different spatial information crucial for model accuracy. These features were considered essential because their low VIF values suggest minimal collinearity, and they contribute independently to the model [43]. In addition, Pearson correlation analysis [44] was used to evaluate the spatial interconnectivity of predictor relationships.

The results indicate that, except for the Pearson correlation coefficient of 0.86 between TH and HD (HD reflects the moisture content in the air; high humidity usually reduces the incidence of fire, because a high humidity environment is not conducive to the drying and combustion of combustibles; TD reflects the overall climate conditions, that is, the degree of dryness or wetness; there is often a positive correlation between TD and fire; since TD and HD affect the occurrence of fire from different levels, and the VIF values of TD and HD do not exceed 10, they are retained), most of the variables show a negative correlation, with the maximum positive correlation not exceeding 0.12. The multicollinearity test using the Variance Inflation Factor (VIF) (Supplementary Table S1) indicates that there are no issues with the factors affecting forest fires. The table below shows that the maximum and minimum VIF values for population density (1.0119) and TD (4.6252) are both well below the threshold value of 10.

2.3.2. Application of Machine Learning Models

As mentioned earlier, eight different ML algorithms were used in this paper. In these methods, most combine machine learning models with metaheuristic optimization algorithms to more efficiently and rationally select model parameters. Regarding computational efficiency, although metaheuristic algorithms may be more resource-intensive compared with traditional methods, they offer improvements by balancing exploration and exploitation. This balance allows them to significantly reduce the likelihood of getting trapped in local optima, resulting in more accurate and robust solutions. As metaheuristic optimization is primarily used for parameter selection, the relatively low complexity of these problems makes the computational time incurred acceptable. The ML model used will be introduced below.

PSO-SVM is an integrated method that combines the PSO algorithm SVM model. The primary goal of PSO-SVM is to improve classification or prediction accuracy by optimizing the parameters of the SVM. The core of PSO-SVM lies in using the Particle Swarm Optimization algorithm to automatically select the parameters of the SVM, including the penalty factor and the kernel function parameter [33]. This approach avoids the randomness of manually selecting parameters, thereby enhancing the robustness and classification accuracy of the model [45]. In this study, the number of iterations is set to 50. The number of particles is set to 20. The learning factor is set to 2. These parameters were chosen based on preliminary tests and their performance in balancing convergence speed and accuracy in the model.

RF is an ensemble learning method that improves prediction accuracy and model robustness by constructing multiple decision trees and combining their prediction results [34]. The core idea of the RF algorithm is to use the Bootstrap resampling method to draw multiple samples from the original dataset, build a decision tree for each sample, and then obtain the final prediction by voting or averaging the results [46]. This approach not only enhances the model's prediction accuracy but also exhibits good tolerance to outliers and noise, making it less prone to overfitting. The number of trees is set to 50 in this study following recommendations from existing studies, where a value around 50 has been shown to provide a good trade-off between model complexity and predictive accuracy.

CPO-BP is a nature-inspired optimization algorithm with strong global search capabilities and diversified local search strategies. CPO has been applied and improved in various fields to enhance its optimization performance and adaptability [47]. The CPO algorithm mimics the four defense strategies of the porcupine, including visual intimidation, auditory intimidation, olfactory attack, and physical attack, which correspond to the exploration and exploitation phases of the algorithm. This defense mechanism-based algorithm is considered unique in the field of metaheuristics.

GOOSE-XGBoost is an intelligent optimization algorithm inspired by the resting and foraging behavior of geese [36]. When geese hear any unusual sounds or movements, they emit loud calls to alert the individuals in the flock, ensuring their safety [48]. It is famous for its fast convergence and high solution accuracy. It is an excellent optimization algorithm, which can handle multi-objective optimization problems at the same time.

HO-BP is a metaheuristic optimization algorithm based on the behavior of hippopotamuses, which has been widely applied to optimization problems in recent years. The HO algorithm searches for the optimal solution by simulating the movement, predation, and reproduction behaviors of hippopotamuses [38]. It can solve optimization problems from simple to highly complex, especially in high-dimensional complex problems. This is due to the flexibility of the optimization mechanism.

HLOA-CatBoost is an optimization algorithm inspired by the behavior of horned lizards in nature, primarily used to solve complex optimization problems. HLOA simulates the predation behavior of horned lizards to search for the optimal solution, exhibiting strong global search capabilities and the ability to avoid local optima [37]. As a nature-inspired optimization algorithm. HLOA has demonstrated its powerful optimization ability and adaptability across multiple domains. Through integration and improvement with other algorithms, HLOA has shown outstanding performance in solving complex optimization problems. It is not only suitable for high-dimensional optimization problems, but also can effectively deal with discrete and multi-objective optimization problems.

BFO-KELM is an optimization algorithm inspired by the foraging behavior of bacteria. This algorithm solves optimization problems by simulating bacterial chemotaxis, reproduction, and migration behaviors [39]. The BFO algorithm has been applied in various fields, including power system control, feature selection, and combinatorial optimization. By mimicking the foraging behavior of bacteria, the BFO algorithm possesses strong global search capabilities and good convergence properties, making it suitable for a wide range of complex optimization problems. It has demonstrated good adaptability in high-dimensional optimization, multi-objective optimization, and nonlinear problems.

PO-BP is an optimization algorithm inspired by the foraging behavior of parrot flocks in nature. This algorithm solves optimization problems by simulating the foraging behavior of parrots, featuring adaptive weights and learning factors, which effectively avoid the problems of local optima and low precision commonly encountered in traditional Particle Swarm Optimization algorithms [40]. It has a fast convergence speed and high stability.

2.4. Verification Analysis and Weighting

This paper compared eight different ML classification results. Common metrics for binary classification problems were used, including accuracy [49], kappa coefficient [50], area under the curve (AUC) [51], sensitivity [52], specificity, positive predictive value (PPV), and negative predictive value (NPV) [53]. The final training results are shown in Table 3. To generate a fire risk map, this paper selects the training model based on the training results. Due to the good performance of HLOA, RF, and GOOSE optimization methods (with accuracy values exceeding 0.8; kappa values of 0.49, 0.50, and 0.50; and AUC values exceeding 0.7), these three methods were used for plotting.

Table 3. Verification table.

Method	Accuracy	Kappa	Sensitivity	Specificity	PPV	NPV	AUC
PSO-SVM	0.8268	0.9775	0.3448	0.8510	0.8000	0.9846	0.6115
CPO-BP	0.2537	0.0000	0.0000	1.0000	0.0000	0.2537	0.5000
HLOA-CatBoost	0.8427	0.4936	0.9231	0.5424	0.8831	0.6531	0.7327
HO-BP	0.7393	0.3507	0.9130	0.3973	0.8112	0.6170	0.6551
BFO-KELM	0.2397	0.0044	0.0093	1.0000	1.0000	0.2347	0.5046
PO-BP	0.7464	0.0000	1.0000	0.0000	0.9491	0.0000	0.5000
RF	0.8050	0.5021	0.8649	0.6724	0.9100	0.5652	0.7686
GOOSE-XGBoost	0.8071	0.5081	0.8589	0.4918	0.8714	0.7692	0.7253

For the fire risk results of the three optimization methods ultimately selected, we employ an attention mechanism [54] to fuse the results from three selected models. The attention mechanism, originally derived from neuroscience, has achieved significant success in the field of deep learning, particularly in natural language processing (NLP). The core idea behind the attention mechanism is that, when processing an input sequence, not all elements are given equal importance. Instead, the model assigns different weights to various parts of the input, allowing it to focus on the more relevant components. The weights for each method's results are assigned based on the calculation of the Mean Squared Error (MSE) for each model by Formula (1) (where a represents the total MSE of the three models, i represents the model number, and w represents the weight); the smaller the error, the greater the weight. In the weight allocation, we opted to determine the final weights of each model based on the MSE, without incorporating other evaluation metrics such as AUC or sensitivity. The primary reason for this decision is that other metrics, AUC and sensitivity, have already been extensively utilized during the initial model screening and evaluation phase. Reintroducing these metrics in the weight allocation stage would lead to a situation where the weighted model's AUC and sensitivity values would be significantly higher than the original values of the individual models in cross-validation, regardless of the actual performance. Therefore, selecting MSE as the sole weighting criterion is helpful in avoiding evaluation bias caused by the repeated application of multiple metrics, thereby enhancing reliability of the results. The final weights assigned to these three methods (RF, GOOSE, HLOA) are 0.32, 0.33, and 0.34. The weighted result is obtained from Formula (2) [55].

$$w_i = \frac{(mse_{total} - mse_i)}{mse_{total}} \quad (1)$$

$$result = result_1 * w_1 + result_2 * w_2 + result_3 * w_3 \quad (2)$$

3. Results and Discussion

3.1. Factors Importance Analysis

The RF method is commonly used for feature importance analysis [49]. When performing feature importance analysis with RF, the Variable Importance Measure is typically used to assess the impact of each variable on the model's prediction results. There are two main methods for calculating variable importance in Random Forest: one is through permutation of variable values, and the other is by using the Gini index. Since the permutation method has better unbiased performance, it is more commonly used for variable selection. Additionally, RF can measure variable importance through metrics including minimum depth, frequency of variable splits at the root node, and node purity. These methods help identify key variables that contribute significantly to the model's predictions, thereby reducing the dimensionality of the input space in high-dimensional datasets and improving computational efficiency.

The most influential factor for forest fire occurrence is TD, followed by ARP and HD, which correspond to daily average temperature, daily average air pressure, and dew point temperature, respectively. Therefore, the TD value reflects the potential threat of a fire, and this value contributes most significantly to forest fires in the Yongqiao District. The higher the TD, the faster the evaporation rate of water in the combustibles, and the items become dry, which is prone to fire. The higher the DOP, the more human activities, and the more prone to fire (Figure 4). These two factors may interact synergistically; for instance, in areas with both high temperature and high population density, the combination could create a heightened risk of fire occurrence due to both the physical drying of combustibles and increased human activity in vulnerable areas. Relevant management departments can incorporate this indicator to enhance future fire prevention planning.

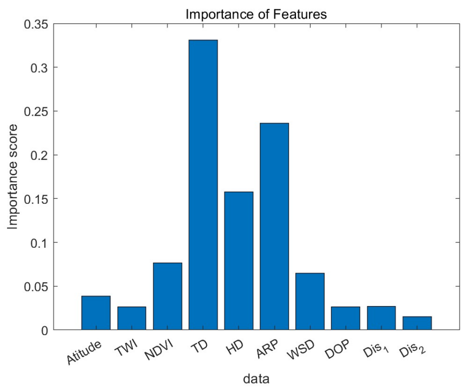
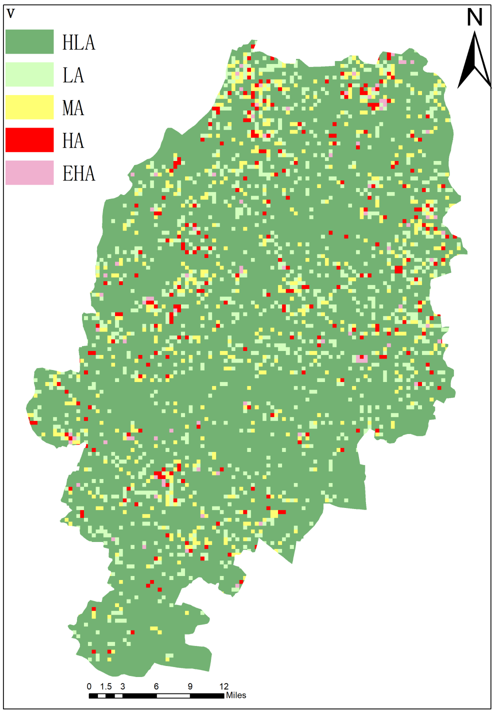


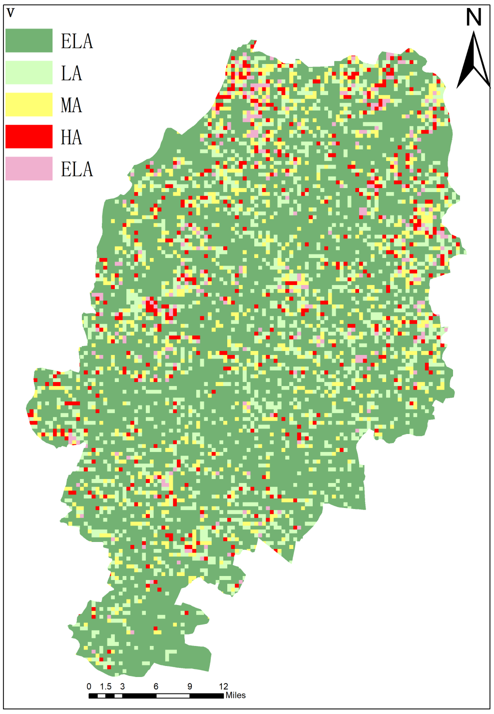
Figure 4. Factors' importance.

3.2. Fire Risk Map

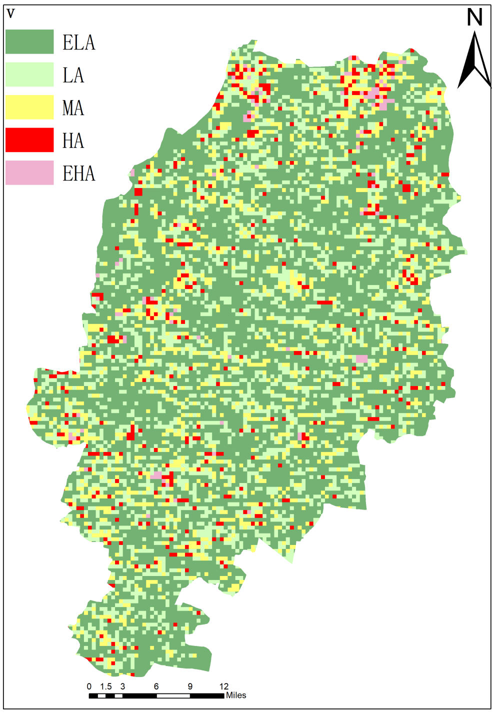
Based on the predicted probability of each grid cell, this paper gridizes the study area and applies the natural fracture method to classify the fire risk level into five categories: ELA (extremely low-risk area), LA (low-risk area), MA (medium-risk area), HA (high-risk area), and EHA (extremely high-risk area). Figure 5 shows the fire risk map of the study area of three selected machine learning methods. After weighting the three selected methods with Formula (1), the integrated fire risk map is consistent with the three selected methods (Figure 5d), with areas of higher fire risk concentrated in the northwest, northeast, and south of the study area.



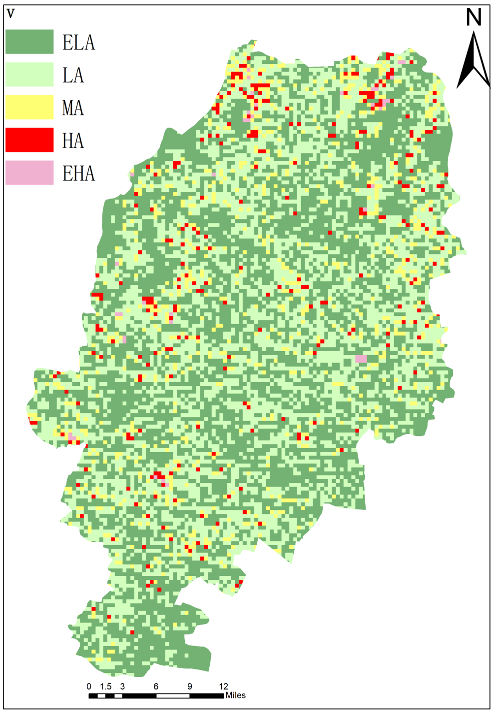
(a) GOOSE



(b) HLOA



(c) RF



(d) Fire Risk Map

Figure 5. Fire risk map of three selected methods and the final fire risk map.

To validate the performance of the weighted model, cross-validation was performed using previous data, the accuracy and AUC values of the weighted model were higher than those of the three selected models (Table 4) .

Table 4. Cross-validation.

Method	Accuracy	Kappa	Sensitivity	Specificity	PPV	NPV	AUC
HLOA-CatBoost	0.8427	0.4936	0.9231	0.5424	0.8831	0.6531	0.7327
RF	0.8050	0.5021	0.8649	0.6724	0.9100	0.5652	0.7686
GOOSE-XGBoost	0.8071	0.5081	0.8589	0.4918	0.8714	0.7692	0.7253
Integrated	0.8602	0.5620	0.9234	0.6207	0.9031	0.6792	0.7720

Low-risk areas and extremely low-risk areas occupy nearly 90% of the area (Figure 6), indicating that during the analysis period, the fire risk in most areas is at a controllable level, making them suitable for regular fire prevention management measures. Of course, in the forest fire early warning model, special attention should be given to the high-risk and extremely high-risk areas, which together occupy nearly 3% of the area. Therefore, during the forest fire-prone months of May and June, the forest fire prevention and control efforts in this region should focus on these high-risk areas and implement targeted preventive and response measures.

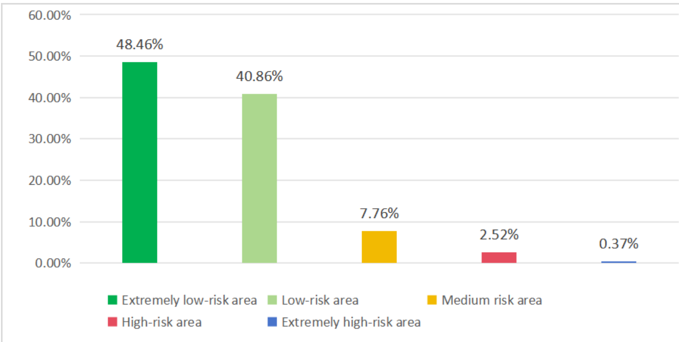


Figure 6. Proportion of fire-prone areas in the study area.

3.3. Discussion

Moran's I quantifies spatial autocorrelation, which is crucial for phenomena exhibiting geographic dependence, such as forest fires. Future work will supplement the analysis by including geospatial autocorrelation analysis to account for spatial dependencies between observations. This will enhance the rigor of our spatial modeling and provide more accurate predictions by considering the spatial structure of the data. In addition, other dimensionality reduction techniques could be considered for feature selection. For instance, Principal Component Analysis (PCA) could be used to transform correlated features into a set of uncorrelated principal components [56]. Alternatively, Least Absolute Shrinkage and Selection Operator (LASSO) regression could be employed to perform feature selection and reduce the impact of irrelevant or redundant features by penalizing large coefficients [57]. Further exploration of these techniques is encouraged in future studies. Stacking and blending tend to use the outputs of multiple base models to generate new feature sets, which improves the generalization ability of the model, while weighting emphasizes the increasing influence on the model with good performance. We chose weighting because we used this framework in the specific location of the Yongqiao District. We will use stacking or blending in future work. Moreover, the generalizability of the proposed method to different climatic conditions remains an open issue. Given that fire behavior is highly sensitive to environmental factors, future work should evaluate the model's performance across diverse climatic regions, ensuring its adaptability in other studied area. Additionally, applying this approach to larger datasets with varying spatial and temporal scales would help assess its scalability and robustness.

Though the RF-based feature importance is in line with expectations, its validity could be further reinforced by using more interpretable methods. In future work, SHAP (Shapley Additive Explanations) values will be considered to better understand feature contributions and interactions [58], and use permutation importance as a complementary technique to assess the stability and robustness of the feature ranking. Furthermore, spatial validation is critical for the fire risk maps. While the proposed method demonstrates effectiveness in a specific region, its transferability to other geographic contexts remains uncertain. In addition to comparing the maps with historical fire occurrences, future study will consider incorporating ground truth data, such as fire reports, satellite imagery, or data from local fire departments, to further validate the predictions. The growing potential of deep learning techniques, such as Convolutional Neural Networks (CNNs) for spatial data processing [59] and Long Short-Term Memory networks (LSTMs) for spatiotemporal modeling [40], is worthy of consideration. These methods have shown promising results in similar applications, particularly in handling complex, high-dimensional data. Moving forward, it is worthy to explore the integration of these techniques into our framework to assess their ability to improve prediction accuracy and enhance the robustness of fire risk assessment models.

4. Conclusions

Fire risk map is an effective method for quantifying regional fire risk. However, most existing forest fire risk mapping studies use a single machine learning model, and different models may have different performance in the same spatial environment. Therefore, it is crucial to evaluate the differences in model performance and integrate the predictions of different models. We analyzed the factors that affect the occurrence of forest fires and evaluated the results of different models predicting forest fires. Then, we used weighting to integrate the results of different models. We used eight machine learning models for training. The results obtained by different machine learning methods in the same region show different trends. Therefore, based on the different performance of different machine learning methods in obtaining different results, we choose the model with better performance and use weighted methods to integrate different methods together for graph generation. The AUC and accuracy values obtained through cross-validation indicate that the weighted performance is superior to the original performance. The study area has many valuable natural resources and tourist attractions, but there is no fire risk map. The obtained fire risk framework can be used to draw a fire risk map. This method provides valuable insights for solving future fire problems. While this study advances forest fire risk modeling, several limitations warrant consideration. The framework's adaptability to divergent climatic regimes and scalability on expansive geospatial datasets require rigorous verification. Subsequent research should investigate spatial autocorrelation effects, integrate hybrid deep learning architectures, validate the results with ground truth data and historical fire occurrences, and test scalability with larger datasets spanning broader spatial and temporal scales ultimately strengthening the model's ecological representativeness and operational applicability.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/f16020329/s1, Table S1: VIF Table; Figure S1: The base map of factors; Figure S2: The base map of climate factors; Figure S3: Pearson correlation results.

Author Contributions: Conceptualization, C.W.; methodology, H.L.; software, Y.X.; resources, F.Z.; data curation, C.W.; writing—original draft preparation, Y.X.; writing—review and editing, H.L.; visualization, C.W.; supervision, F.Z.; All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported in part by the Start-up Fund for New Talented Researchers of Nanjing Vocational University of Industry Technology (Grant No. YK22-05-01).

Data Availability Statement: Data are contained within the article.

Conflicts of Interest: The authors declare no conflicts of interest.

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